

3.6.1 Electrical current

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 1			
Electric current is a flow of electrical charge. The size of the electric current is the rate of flow of electrical charge.	Electrical current is measured in amps using an ammeter.	6.2.1.2	4.7.2.1
The current through a component depends on both the resistance of the component and the voltage across the component. The greater the resistance of the component the smaller the current for a given voltage across the component.	Voltage is measured in volts using a voltmeter. The term 'potential difference' will not be used.	6.2.1.3	4.7.2.2
The resistance of a component is a measure of how difficult it is for an electric current to pass through it.	Calculations of resistance will not be required, nor will the units of resistance. Suggested activity for TDA Investigate which materials are the best electrical conductors.	6.2.1.3	4.7.2.2
Outcome 2			

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
A complete circuit is necessary for a current to flow. Cells and batteries supply current that always passes in the same direction. This is called direct current (d.c.).	No distinction between series and parallel circuits needs to be made. The use of conventional circuit diagram symbols will not be required.	6.2.2	4.7.2.5
An alternating current (a.c.) is one that changes direction. Mains electricity is an a.c. supply. In the UK it has a frequency of 50 Hz and is 230 V.		6.2.3.1	4.7.2.5